BoMu: A Resistively-Loaded NMOS Differential Pair Design & Sizing (SKY130A, TinyTapeout)

Mathew Hanson

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1 Introduction

`BoMu` is a single-stage differential amplifier built from a resistively-loaded NMOS source-coupled pair, designed for the SKY130A PDK and open-source TinyTapeout flow. Two matched NMOS devices (M_1, M_2) share a tail sink of $2I_D$, and each drain is pulled to the $1.2V$ supply through a load resistor R_D. The amplifier output is taken differentially across the drains, $v_{od} = v(\text{voutp}) - v(\text{voutn})$. A floating differential source $\pm v_{sd}/2$ on a common-mode V_{SC} drives the gates through 100Ω series resistors.

The driving bandwidth constraint is that each output drives an I/O pad of fixed $C_L = 5pF$. The tail sink is realised on-chip as a simple current mirror (Section 8.8) with a $200mV$ compliance budget. The flow is fully open-source (`xschem`, `ngspice`, `magic`/`klayout`); devices are sized by a g_m/I_D methodology [1] from measured current-density and speed data rather than an assumed square-law constant, and the small-signal results are standard [2].

This report documents the adopted $W = 80\,\mu m$ design and signs it off at the `tt` corner (with the native device-mismatch Monte Carlo of Section 10). The design is placed explicitly on the g_m/I_D area–power trade (Sections 8.3–8.4), where it sits at the optimum of a power-weighted cost. The common-centroid `magic` layout is then drawn, closed LVS/DRC clean, and re-simulated against its extracted parasitics (Section 11), which confirms both targets survive the layout.

1.1 Net identities

With the sign convention M_1 gate = `vinp` → drain = `voutn`, and M_2 gate = `vinn` → drain = `voutp`, the differential output $v_{od} = v(\text{voutp}) - v(\text{voutn})$ has positive gain with respect to the differential input $v_{id} = \text{vinp} - \text{vinn}$. The two half-circuits are identical by matching, so the analysis is done on a single common-source half-circuit and the differential result follows directly.

2 Specifications

The target specifications, and recommended design parameters, are collected in Table 1.

Table 1: Design targets and recommended parameters.

Parameter	**Value**	**Units**
Small-signal bandwidth f_{-3dB}	150	MHz
Differential DC gain A_{v0}	10	dB
Differential output swing v_{od}	400	mV_{pp}
Channel length L	0.15	μm
Supply V_{DD}	1.2	V (fixed)
Pad load C_L (each output)	5	pF (fixed)
Tail-sink compliance budget	200	mV (fixed)

3 First-Order Hand Design

The design proceeds as follows: the fixed pad capacitance and the bandwidth target set the load resistor, the gain target then sets the required transconductance, and a chosen inversion level (g_m/I_D) fixes the

bias current.

Load resistor from bandwidth. The dominant pole sits at the output node, formed by the load resistance working into the fixed C_L. Ignoring the device output resistance for the moment,

$$R_D \;=\; \frac{1}{2\pi f_{-3dB} C_L} \;=\; \frac{1}{2\pi(150\text{MHz})(5\text{pF})} \;\approx\; 212\Omega. \tag{1}$$

Transconductance from gain. The differential (half-circuit) gain of a resistively-loaded pair is

$$A_{v0} \;=\; g_m(R_D \| r_o) \;\xrightarrow{r_o \gg R_D}\; g_m R_D. \tag{2}$$

With $A_{v0} = 10\text{dB} = 3.162\text{V/V}$ and Eq. (1),

$$g_m \;=\; \frac{A_{v0}}{R_D} \;=\; \frac{3.162}{212\Omega} \;\approx\; 14.9\text{mS}. \tag{3}$$

Bias current from g_m/I_D. Choosing a moderate-inversion $g_m/I_D = 15\text{V}^{-1}$ (a balance of device area, r_o, and speed at this minimum length) fixes the per-branch current and the tail current,

$$I_D \;=\; \frac{g_m}{(g_m/I_D)} \;=\; \frac{14.9\,\text{mS}}{15} \;\approx\; 0.99\text{mA}, \qquad I_{\text{tail}} = 2I_D \approx 2\text{mA}. \tag{4}$$

Swing and headroom. A $400mV_{pp}$ differential swing corresponds to a single-ended peak of $v_{od}/4 = 100mV$ about each drain's DC level $V_D = V_{DD} - I_D R_D \approx 0.99\,V$. Two conditions must hold: the bias drop must be large enough to source the swing, $I_D R_D > 100mV$ ($R_D = 212mV$ is ample), and the device must remain saturated at the bottom of the swing, $V_{D,\min} - (V_S + V_{ov}) > 0$, where V_S is the tail-node voltage held at or above the $200\,mV$ compliance budget. Both are comfortably satisfied because R_D is small and the supply headroom is large.

4 g_m/I_D Sizing Flow

Equations (1)–(3) fix R_D and the required g_m. The g_m/I_D method then turns the chosen inversion level into a physical width. A unit device is characterised over V_{GS} at fixed (V_{SB}, V_{DS}) to build a lookup grid returning, for any g_m/I_D, the current density J_D ($=I_D/W$), the intrinsic gain $g_m r_o$, and the transit frequency f_T. The width then follows as

$$W \;=\; \frac{I_D}{J_D(g_m/I_D,\, V_{SB},\, V_{DS})}. \tag{5}$$

This is implemented in `sim/sizing.py (worksheet)`, which references the `extern/sky130A-gmid` lookup grid (a git submodule). The script takes a `DESIGN` dict (targets + the two free knobs, g_m/I_D and V_S), prints the worksheet of Table 2, and with `--spice` generates a differential-pair deck, runs `ngspice`, and reports predicted-versus-SPICE side by side (Table 3).

Table 2: Sizing worksheet at example operating point ($g_m/I_D = 15$, $V_S = 0.30\,V$, $L = 0.15\,\mu m$, SKY130A `nfet_01v8_lvt`).

Quantity	**Value**	**Units**
R_D (from BW)	212	Ω
g_m required (from gain)	14.9	mS
I_D / I_{tail}	0.99 / 1.99	mA
$V_{ov} = 2/(g_m/I_D)$	0.133	V
current density J_D	14.6	$\mu A/\mu m$
width W (per device)	68.0	μm
output resistance r_o	941	Ω
intrinsic gain $g_m r_o$	14.0	—
transit frequency f_T	51.3	GHz
bias nodes V_D / V_{DS}	0.989 / 0.689	V
power $V_{DD}\cdot I_{\text{tail}}$	2.38	mW

5 The r_o-Limited Gain

Equation (2) assumed $r_o \gg R_D$, but at minimum length this does not hold. The lookup returns $r_o \approx 941\Omega$ at $g_m/I_D = 15$, so the effective load is

$$R_L \;=\; R_D \| r_o \;=\; 212 \| 941 \;\approx\; 173\Omega, \tag{6}$$

and the realised gain and bandwidth become

$$A_{v0} = g_m R_L \approx 2.58 \text{ V/V } (8.23\text{dB}), \qquad f_{-3dB} = \frac{1}{2\pi R_L C_L} \approx 184\text{MHz}. \tag{7}$$

Because R_D is pinned at 212 Ω by the fixed C_L and the bandwidth target (Eq. (1)), and r_o at this length is comparable to R_D, the gain + BW targets are competing: the finite r_o pulls the gain down to $\sim 8.2dB$ while pushing the bandwidth up past $150MHz$. A single resistively-loaded stage at $L = 0.15\mu m$ cannot simultaneously meet $10dB$ and $150MHz$ for the selected R_D at the baseline current.

Figure 1 shows this: sweeping the inversion level with g_m held at the gain-target value ($14.9\,mS$), the realised gain plateaus near $8.3dB$ and never reaches $10\,dB$. At $L = 0.15\,\mu m$ the intrinsic $g_m r_o$ is simply too small for r_o to dominate R_D, regardless of bias. The resolution (Section 8.6) is a small increase in g_m — i.e. a little more current — which lifts the realised gain–bandwidth product enough to clear both targets.

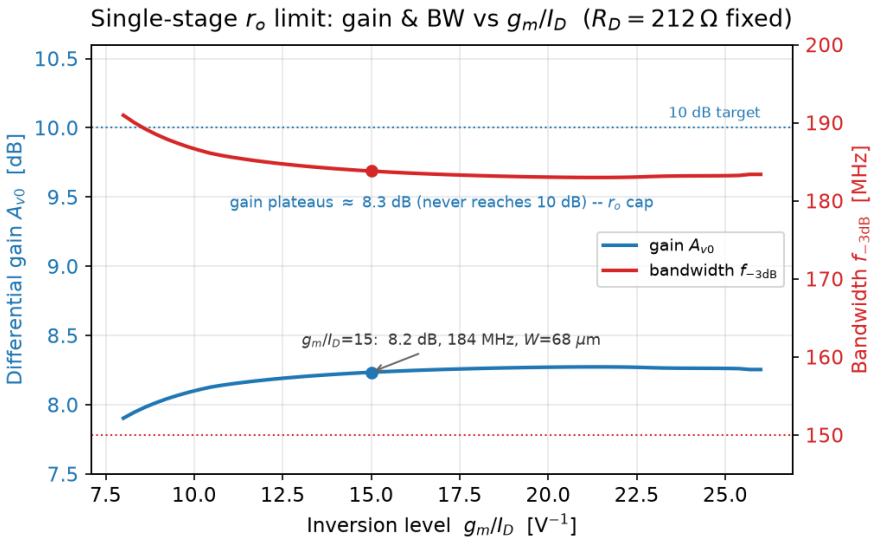


Figure 1: Gain and bandwidth versus inversion level g_m/I_D, with g_m fixed at $14.9mS$ and $R_D = 212\Omega$. The gain plateaus near $8.3dB$.

6 SPICE Verification (SKY130A, tt corner)

The self-contained differential AC testbench `sim/tb_ac.spice` (floating $\pm v_{sd}/2$ drive, 100Ω gate resistors, on-chip $2I_D$ tail mirror, $5pF$ pad caps) exercises the design under `ngspice` with the SKY130A `tt` models. The hand-calculations track SPICE closely as shown in Table 3. Particularly, the $R_D \| r_o$ gain estimate of Section 5 reproduces the simulated gain to within $0.01dB$, confirming the r_o-limited result is real.

Table 3: Predicted (hand-calc) versus SPICE at the baseline sized operating point ($W = 68\mu m$, $g_m/I_D = 15$).

Quantity	Predicted	SPICE	Units
Tail current I_{tail}	1987	1987	μA
g_m per device	14.9	14.85	mS
Tail-node voltage V_S	0.30	0.205	V
Differential gain A_{v0}	8.23	8.24	dB
Bandwidth f_{-3dB}	183.8	177.8	MHz

The simulated response (Figure 2) is a flat $\sim 8.2dB$ passband rolling off at the single output pole, below the $10dB$ gain target and above the $150MHz$ bandwidth target — the r_o-limited baseline that the gain-recovery of Section 8.6 corrects. The tail node settles at $205mV$ (just above the $200mV$) and the drain DC operating point of $0.988V$ matches $V_{DD} - I_D R_D$ exactly.

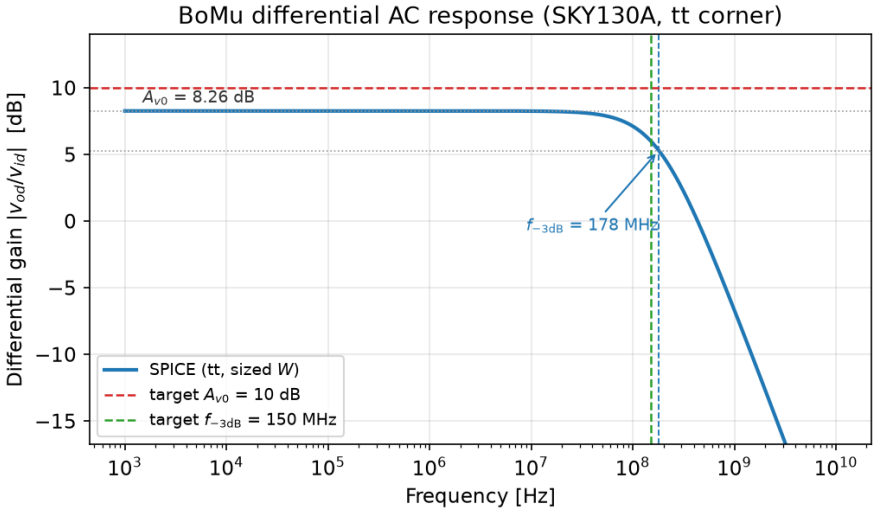


Figure 2: Differential AC response from `sim/tb_ac.spice` (SKY130A, `tt`, baseline $W = 68\mu m$).

7 Design Bookkeeping: Predicted vs. SPICE Across the Flow

Table 3 spot-checked five quantities; this section carries the *entire* g_m/I_D sizing flow as a single Murmann-style design-cockpit table (Table 4), each row quoting the hand/grid prediction against SPICE with a percentage error, at the baseline $W = 68\mu m$ sizing point the method produces. The table is generated directly by `sim/sizing.py cockpit` (console) and `--latex` (this tabular), so it stays in sync with the sizing script. Of the sixteen rows carried predicted-vs-SPICE the mean absolute error is 1.1% and the worst is 6.5%.

Table 4: Design-cockpit bookkeeping at the baseline sized operating point ($W = 68\mu m$, $g_m/I_D = 15$, `tt`, $1.2V$, 27°C). Generated by `sim/sizing.py cockpit --latex`.

Parameter	g_m/I_D pred.	SPICE	%err	note
R_D [Ω]	212.2	–	–	set by BW & C_L (ideal load)
g_m/I_D [1/V]	15	14.95	-0.3	inversion level (chosen)
I_D/branch [μA]	993.5	993.5	+0.0	gm_req/(gm/Id)
I_{tail} [mA]	1.987	1.987	+0.0	= 2*Id (VDD current)
V_{ov} [mV]	133.3	142	+6.5	2/(gm/Id) vs Vgs-Vth
J_D [μA/μm]	14.61	14.61	+0.0	current density (grid)
W [μm]	68.02	–	–	= Id/Jd (then rounded)
g_m/device [mS]	14.9	14.85	-0.3	= Av0/R_D (ideal)
r_o [kΩ]	0.9415	0.9598	+1.9	output resistance
$g_m r_o$ [V/V]	14.03	14.25	+1.6	intrinsic gain
f_T [GHz]	51.32	51.03	-0.6	= gm/(2pi.Cgg)
C_{gg} [fF]	46.21	46.32	+0.2	gate cap
R_L=$R_D \| r_o$ [Ω]	173.2	173.8	+0.4	real load
A_{v0} [dB]	8.234	8.235	+0.0	= gm.R_L
f_{-3dB} [MHz]	183.8	177.8	-3.3	= 1/(2pi.R_L.C_L)
GBW [MHz]	474.3	458.9	-3.2	gain-bandwidth
P_{tot} [mW]	2.384	2.384	+0.0	= VDD.I_tail
V_D (drain) [V]	0.9892	0.9892	-0.0	drain DC
V_{tail} [V]	–	0.2047	–	ideal tail src; ≥0.2 V compliance met
SE swing peak [mV]	100	–	–	= v_od/4 (headroom check)

The table shows two things. The current-domain rows (the grid density J_D, hence W, and I_D, g_m, f_T, C_{gg}, r_o, $g_m r_o$) all land inside 2%: the g_m/I_D grid predicts the sized device from the targets with

essentially no iteration. The one visible gap is V_{ov} (+6.5%): the $V_{ov} \approx 2/(g_m/I_D)$ rule-of-thumb underestimates the true overdrive in moderate inversion, but because W is taken from the grid current density J_D (not from V_{ov}), that approximation does not propagate into the sizing. The output rows then confirm the baseline sits where the flow places it — $A_{v0} = 8.2\,\text{dB}$ and $f_{-3\text{dB}} = 184\,\text{MHz}$ — i.e. comfortably past the bandwidth target but short of the 10 dB gain, the r_o-limited shortfall of Section 5 that the small g_m increase of Section 8.6 then recovers to reach the adopted $W = 80\mu m$ design.

8 Design-Space Sweeps and the Gain–Bandwidth Limit

Section 5 showed that at the chosen $R_D = 212\Omega$, the finite r_o leaves the gain short and the bandwidth long. This section sweeps R_D and L to map the achievable space.

8.1 R_D: trading gain for bandwidth at constant $g_m/(2\pi C_L)$

For a single output pole the gain–bandwidth product is

$$A_{v0} \cdot f_{-3dB} = g_m(R_D \| r_o) \cdot \frac{1}{2\pi(R_D \| r_o)C_L} = \frac{g_m}{2\pi C_L}, \tag{8}$$

With $g_m = 14.9mS$ and $C_L = 5pF$ this ideal product is $474MHz$. Sweeping R_D (Figure 3) therefore slides the design along this fixed hyperbola. A larger R_D buys gain at the cost of bandwidth. This sweep is at constant power, since R_D draws no bias current. It is also, by itself, not a route to the gain target: once the two targets pin the product, no R_D clears both at the baseline g_m (quantified in Section 8.5).

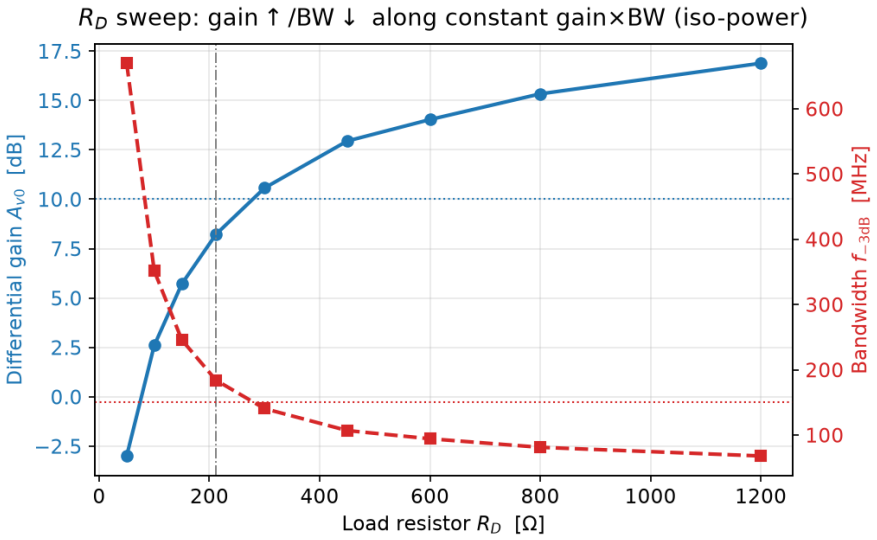


Figure 3: R_D sweep (iso-power $2.38mW$): gain rises and bandwidth falls along the constant $g_m/(2\pi C_L)$ product. Dotted lines mark the $10dB$ and $150MHz$ targets.

Channel length was also swept in an attempt to recover additional gain. Longer L raises r_o, so $R_D \| r_o \to R_D$ and the gain climbs toward the ideal $g_m R_D$ at constant power (the current is unchanged, $W \propto L$ to hold the inversion level). The recovery is modest and saturating: $8.2dB$ at $L = 0.15\mu m$ up to only $\approx 9.1dB$ by $L = 0.6\mu m$, which was still short of the $10dB$ target confirming the r_o-limited ceiling of Section 5. Length trades area for a fraction of the missing gain but does not, on its own, reach the target.

8.2 The design space: bandwidth versus power at fixed gain and swing

The R_D sweep moved along one iso-power hyperbola; the next question is how much bandwidth is available once power is allowed to vary, subject to the fixed $10dB$ gain and $400mVpp$ swing. For the

single output pole, eliminating the load between $A_{v0} = g_m(R_D \| r_o)$ and $f_{-3dB} = 1/(2\pi(R_D \| r_o)C_L)$ gives the load-independent relation

$$f_{-3dB} \;=\; \frac{g_m}{2\pi A_{v0} C_L} \;=\; \frac{(g_m/I_D)\, I_D}{2\pi A_{v0} C_L}, \tag{9}$$

so at fixed A_{v0} and the fixed $5pF$ pad the bandwidth is set *only* by g_m. Since $P = V_{DD}\, I_{tail} = 2V_{DD} I_D$, bandwidth therefore scales *linearly with power*, and the slope of that line is the g_m/I_D efficiency. Figure 4 maps this plane: each line is one inversion level, swept over current.

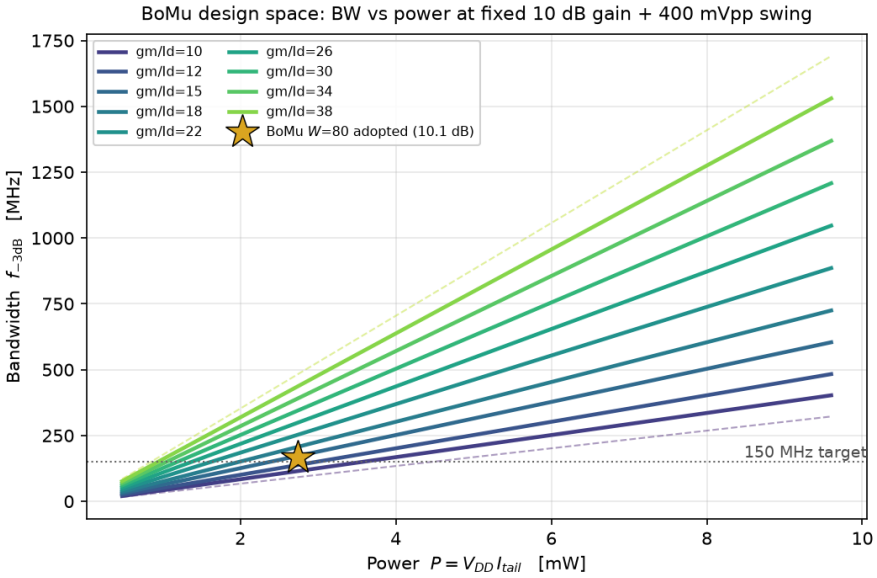


Figure 4: Design-space frontier: achievable f_{-3dB} versus power at fixed $10dB$ gain and $400mVpp$ swing, one line per g_m/I_D (slope $\propto g_m/I_D$). Dashed segments are infeasible; the gold star is the adopted $W = 80\mu m$ design on the $g_m/I_D = 15$ ray ($2.74mW$, $10.1dB$, $168MHz$).

Note first that feasibility is essentially current-independent: the headroom and swing walls are conditions on g_m/I_D, not on power, so each inversion level is feasible (or not) across the whole power axis. The nominal feasible window is wide, $g_m/I_D \approx 9$ to 41: below ≈ 9 the drain headroom fails (the overdrive is large and, with $g_m r_o$ low, the gain-setting R_D grows large enough to collapse the drain), and above ≈ 41 the R_D drop falls below the $100mV$ single-ended swing. Within that window the only way to raise bandwidth at fixed power is to raise g_m/I_D: at the $2.38mW$ baseline, moving from $g_m/I_D = 15$ ($150MHz$) up toward the swing wall would, on paper, reach $\sim 380MHz$ at the same power.

That wide window is misleading, though. BoMu holds a moderate $g_m/I_D = 15$ ($V_{ov} \approx 130mV$) rather than chasing the swing wall for robustness: a higher g_m/I_D means a smaller V_{ov} and less $g_m r_o$ margin, so the stage becomes progressively more sensitive to the device spread the sizing is supposed to tolerate, and the moderate inversion keeps that headroom. The practical lever for more bandwidth is therefore power (the slope of Eq. (9)), not a freer inversion level — and, symmetrically, the only way to cut power at these specs is to cut g_m, which gives back bandwidth one-for-one.

Where the design sits, and the minimum it costs. Figure 5 recasts the same plane with gain on the vertical axis, which makes the hard-target minimum explicit. With $150MHz$ pinning the load at $R_L \leq 212\Omega$, the gain at the bandwidth edge is $A_{v0} = g_m R_L = (g_m/I_D)\, I_D \cdot 212\Omega$, so each inversion level is a ray rising with power (slope $\propto g_m/I_D$) until it saturates at the $g_m r_o$ ceiling. The power at which a ray crosses $10dB$ is the minimum to meet *both* targets at that g_m/I_D. On the $g_m/I_D = 15$ ray adopted here that floor is $2.38mW$; the adopted $W = 80\mu m$ design sits just above it at $2.74mW$ once the gain-bandwidth-wall and parasitic margin (Sections 8.6–8.7) are added. Spending more power moves the design *up* its ray (more gain/BW margin) and spending less moves it down toward the infeasible region — the adopted point is a deliberately small step above the floor.

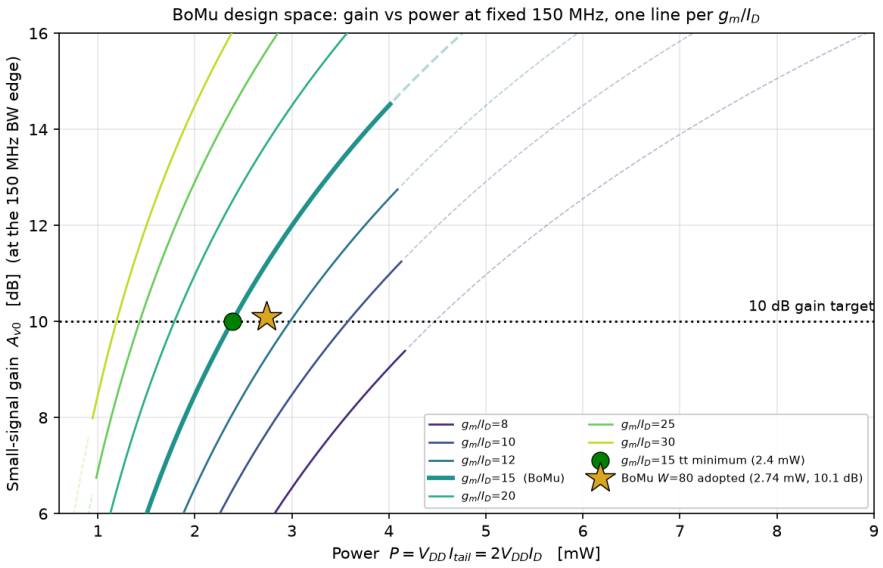


Figure 5: Gain versus power at the fixed $150MHz$ edge, one line per g_m/I_D. On the $g_m/I_D = 15$ ray the $10dB$ target is reachable from $2.38mW$ (green circle); the adopted $W = 80\mu m$ design sits just above it at $2.74mW$. Generated by `sim/plot_gain_space.py`.

8.3 The area–power plane: where each g_m/I_D ray sits

The bandwidth- and gain-versus-power views priced each inversion level in *power*; the third axis that matters on a fixed TinyTapeout tile is *area*. This subsection adds it, so the inversion level can be chosen on the full area–power trade *before* the operating current on the ray is fixed (Sections 8.5–8.7).

Figure 6 projects the $10dB/150MHz$ `tt` criteria onto the area–power plane, one ray per inversion level, in the same house style as the bandwidth- and gain-versus-power frontiers of Section 8.2. At fixed g_m/I_D the pair gate area is linear in power,

$$2WL \;=\; 2L\,\frac{I_D}{J_D} \;=\; \frac{L}{V_{DD}}\,\frac{P}{J_D(g_m/I_D)}, \tag{10}$$

so each g_m/I_D is a ray from the origin whose slope $\propto 1/J_D$ steepens with inversion level (weak inversion spends area fast). The $10dB/150MHz$ spec needs a minimum g_m, hence a minimum current, so each ray is spec-feasible only above an onset power (solid above, dashed below); the locus of those onsets is the `tt` frontier. Along it, strong inversion (low g_m/I_D) gives small area but high power, weak inversion the reverse — area and power trade against each other, bounded below by the drain-headroom wall and above by the output-swing wall of Section 8.2.

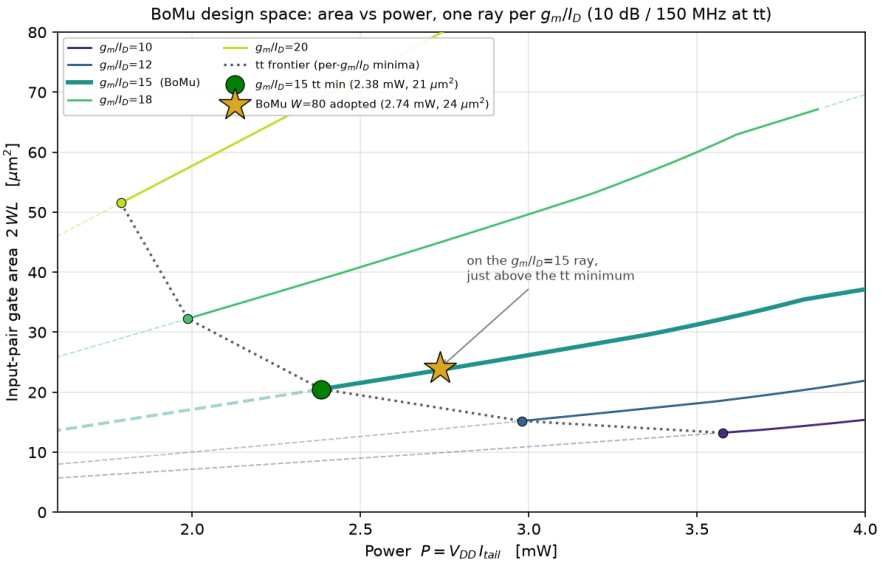


Figure 6: Area–power plane: pair gate area $2WL$ versus power to meet $10dB/150MHz$ at `tt`, one ray per g_m/I_D (solid where spec-feasible, dashed below the onset marker). The adopted $W = 80\mu m$ design (gold star, $2.74mW$, $24\mu m^2$) sits on the $g_m/I_D = 15$ ray, just above its onset minimum (green circle, $2.38mW$, $21\mu m^2$). Generated by `sim/plot_area_power.py`.

The eventual operating point (gold star — the $W = 80\mu m$ design fixed in Sections 8.6–8.7) will sit at the lower-left of this plane, near the minimum area *and* minimum power the `tt` criteria allow at a conservative moderate inversion, on the $g_m/I_D = 15$ ray, a small step above its feasibility onset. Whether $g_m/I_D = 15$ is itself the right ray to be on is settled in the next subsection with a scalar cost.

8.4 The g_m/I_D optimum: a power-weighted cost

Each g_m/I_D ray in Figure 6 meets the spec at a different (area, power) pair, so choosing the inversion level is choosing a point on the area–power frontier. To collapse that two-dimensional trade to a single preference we score each feasible ray with a scalar cost and ask which g_m/I_D minimises it (Figure 7).

Equal weighting is nearly flat. The scale-free equal-weight product $P_{\min} \cdot A$ gives the shallow grey bowl of Figure 7, minimised at $g_m/I_D \approx 12$ but *very* weakly defined (the band $g_m/I_D \approx 10$–15 is within $+10\%$). The flatness is structural: near the square-law limit $J_D \propto V_{ov}^2$ and $g_m/I_D \propto 1/V_{ov}$, so $P \cdot A \propto 1/\big((g_m/I_D)^2 J_D\big)$ is nearly constant — power and area trade almost reciprocally, and their product barely discriminates. The tilt that remains comes from the r_o-gain wall inflating power below the bowl and J_D collapsing (area diverging) above it.

Power-weighting selects $g_m/I_D = 15$. Equal weighting is the wrong model here: it prices a μm^2 and a mW the same, whereas on a fixed TinyTapeout tile area is effectively free and power is the real budget. Valuing power roughly twice the area — the scale-free weighted product $C = P^{0.66} A^{0.33}$ — pulls the optimum *up* in g_m/I_D (toward lower power), from 12 to ≈ 15 (black curve). The same shift lands at $g_m/I_D = 15.0$ under a normalised weighted *sum* $0.66\,\hat{P} + 0.33\,\hat{A}$, so the result is not an artifact of the sum-versus-product choice. BoMu's $g_m/I_D = 15$ is therefore the optimum of this power-weighted cost, and the adopted $W = 80\mu m$ design sits on that ray.

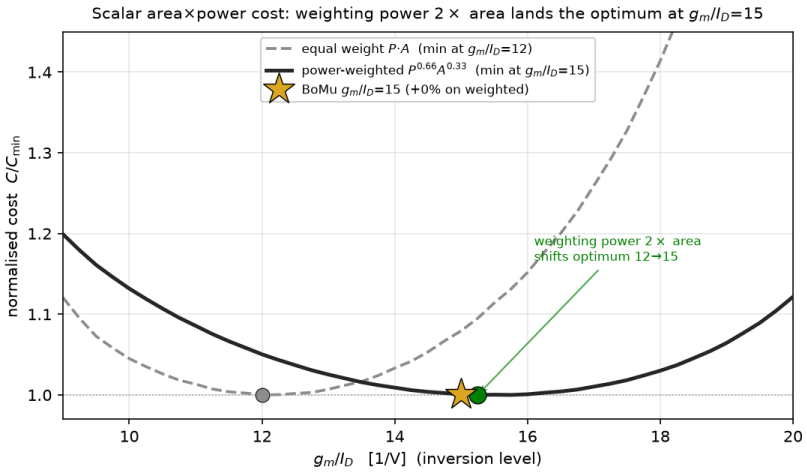


Figure 7: Scalar cost of the area–power trade along the `tt` frontier, each curve normalised to its own minimum. Equal weight $P{\cdot}A$ (grey dashed) minimises at $g_m/I_D \approx 12$; power-weighted $P^{0.66}A^{0.33}$ (black) shifts the optimum to $g_m/I_D \approx 15$, BoMu's choice (star). Generated by `sim/plot_area_power.py`.

Two qualifications are worth stating. The 0.66/0.33 weights are a design *choice* (power-limited tile), not a derivation; a different valuation moves the optimum, and the bowl is shallow enough that anything in $g_m/I_D \approx 13$–17 is near-equivalent, so the cost function corroborates $g_m/I_D = 15$ under a reasonable power-first weighting rather than uniquely forcing it. Independently, a moderate g_m/I_D also carries the larger V_{ov} and $g_m r_o$ headroom that makes the stage tolerant of device spread (Section 8.2), so the power-weighted cost optimum and the robustness-preferred value coincide at 15.

8.5 Sizing the current on the ray: is $R_D \approx 280\,\Omega$ enough?

With $g_m/I_D = 15$ fixed as the cost-optimal ray, the remaining freedom is the operating current on it. The baseline sizing (Section 6: $W = 68\mu m$, $I_{tail} = 1.99mA$) sits at the bottom of that ray; this subsection confirms it cannot reach both targets by re-tuning R_D alone, which motivates the small current increase of Section 8.6. The two targets have product $3.162 \times 150MHz = 474MHz$, which is exactly the ideal GBW of Eq. (8). The specification therefore sits right on the single-stage limit, reachable only if the realised product equals the ideal. It does not: second-order gate/Miller poles pull the SPICE gain–bandwidth product down to $\approx 467MHz$, about 1.5% short. Table 5 brackets R_D in SPICE (`tt`, $L = 0.15$, $W = 68$); note no value of R_D has sufficient GBW product at the baseline g_m.

Table 5: R_D bracket in SPICE.

R_D [Ω]	A_{v0} [dB]	f_{-3dB} [MHz]	GBW [MHz]
220	8.49	175.8	468
250	9.37	158.7	467
265	9.76	**151.7**	467
280	**10.13**	145.4	467
300	10.58	137.9	466

So $R_D \approx 280\Omega$ is a close but still not a feasible solution: the gain passes ($10.13dB$) but the bandwidth is $145MHz$ ($\sim 3\%$ low). Moving to $R_D = 265\Omega$ recovers bandwidth to $152MHz$ but drops the gain to $9.76dB$. The $A_{v0} = 10dB$ crossover lands at $R_D \approx 277\Omega$ with only $f_{-3dB} \approx 148MHz$. The design lands within $\sim$ 2–3% of both but meets neither pair simultaneously, with no margin. Closing the last few percent requires more GBW, which Eq. (8) says comes from a slightly higher g_m (a little more current/power).

8.6 Meeting both specs: a small g_m increase

Because the realised GBW scales with g_m (Eq. (8)), the $\sim 1.5\%$ shortfall is closed by a comparably small increase in g_m. Holding $g_m/I_D = 15$ and $L = 0.15\mu m$, raising g_m from 14.9 to $16.0mS$ (widening the

pair and the tail current) lifts the realised gain–bandwidth product to $\approx 500MHz$, clearing the $474MHz$ the pair requires. Re-centring the load then lands both targets with margin (Table 6).

Table 6: Closing the GBW gap with a g_m increase, SPICE `tt`, $L = 0.15\mu m$, $g_m/I_D = 15$ (ideal tail sink).

g_m [mS]	W [μm]	R_D [Ω]	A_{v0} [dB]	f_{-3dB} [MHz]	P [mW]
14.9 (base)	68	212	8.24	178.0	2.38
15.6	71	272	10.23	150.2	2.50
16.0	73	262	10.15	155.5	2.56
16.0	73	272	10.40	151.1	2.56

The rows above cross the targets near $g_m = 16mS$ ($W \approx 73\mu m$), but the margins there ($+0.15dB$, $+5.5MHz$) are thin. A small additional current buys headroom against the layout parasitics that Section 8.7 addresses, fixing the final operating point.

8.7 Layout-parasitic headroom and the adopted operating point

The $g_m \approx 16mS$ point only marginally clears the bandwidth target, and layout will add output-node capacitance (drain-junction C_{db} plus pad interconnect) on top of the fixed $5pF$. Since $f_{-3dB} = 1/(2\pi R_L C_{tot})$ falls with total capacitance while the DC gain $g_m R_L$ is *unaffected* (a capacitance is an open circuit at DC and carries no bias current), only the bandwidth spec is exposed to parasitics. The remedy is a small current margin: widening the pair to $\mathbf{W = 80\mu m}$ ($I_{tail} = 2.28mA$) on the $R_D \approx 245\Omega$ load lifts the nominal bandwidth well past target with the gain comfortably over $10dB$. At this point the measured response (real `res_high_po` loads and mirror, `tt`) is

$$A_{v0} = 10.1\,\text{dB}, \qquad f_{-3dB} = 168\,\text{MHz}, \qquad P = 2.74\,\text{mW}, \tag{11}$$

so the bandwidth carries $\sim 12\%$ of headroom over $150MHz$ — enough to absorb the few-percent capacitive pole shift a compact layout adds while the gain, which parasitics do not touch, holds its $\sim 0.1dB$ margin. This is the **adopted design**; its sizing is summarised in Table 7.

Table 7: Adopted $W = 80\mu m$ sizing (`ip/bomu.sch`).

Device	**Sizing**	**Role**
M_1, M_2 (pair)	`nfet_01v8_lvt` W=80 μm, L=0.15, nf=8	input diff pair
R_{D1}, R_{D2}	`res_high_po_5p73` L=3.1 μm ($\approx 243\,\Omega$)	drain loads
M_3 (mirror out)	`nfet_01v8` W=80 μm, L=0.5, nf=8	tail sink (N=8)
M_4 (mirror ref)	`nfet_01v8` W=10 μm, L=0.5, nf=1	diode reference
R_{ref}	`res_high_po_2p85` L=5.41 μm ($\approx 745\,\Omega$)	I_{tail} program
I_{tail}	2.28 mA ($V_{SC} = 1.052\,V$)	—

8.8 Tail-current mirror and input common-mode

The ideal tail sink is realised on-chip as a simple two-transistor resistor-programmed NMOS mirror, kept fully internal. A cascode/Wilson mirror needs $V_{GS} + V_{DS,sat} \gtrsim 0.5V$ of compliance, far above the tail budget, so only a simple mirror fits. A raised input common-mode $V_{SC} = 1.052V$ lifts the tail node to $\approx 0.33V$, putting the mirror in clean saturation while the pair retains abundant drain headroom. The $400mV_{pp}$ swing survives (lowest single-ended output $0.82V$, still well above the saturation floor; Section 9).

The mirror uses standard-V_T `nfet_01v8` devices at $L = 0.5\mu m$: output $M_3 = W80\mu m/nf8$, reference $M_4 = W10\mu m/nf1$, ratio $N = 8$, so the reference leg draws only $I_{tail}/8 \approx 0.285mA$ through $R_{ref} \approx 745\Omega$ from V_{DD}. An integrated AC simulation (`sim/tb_ac.spice`) with the real mirror replacing the ideal sink closes the operating point and leaves the differential spec essentially untouched (Table 8): at $R_{ref} = 745\Omega$ the tail settles at $0.333V$, $I_{tail} = 2.28mA$, $A_{v0} = 10.19dB$, $f_{-3dB} = 167MHz$ (ideal R_D; the real `res_high_po` load trims this to the $10.1dB/168MHz$ of Section 8.7). The gain degrades by only $\sim 0.03dB$ from the ideal source, as the mirror's finite r_{out} lands on the common-mode path, not the differential one. The programming resistor sets I_{tail}, so R_{ref} is the single knob for the gain/BW trim. The mirror

is folded into `ip/bomu.sch` (M_3/M_4, R_{ref}); the complete schematic — pair, dummy-flanked loads and tail mirror — is shown in Figure 8.

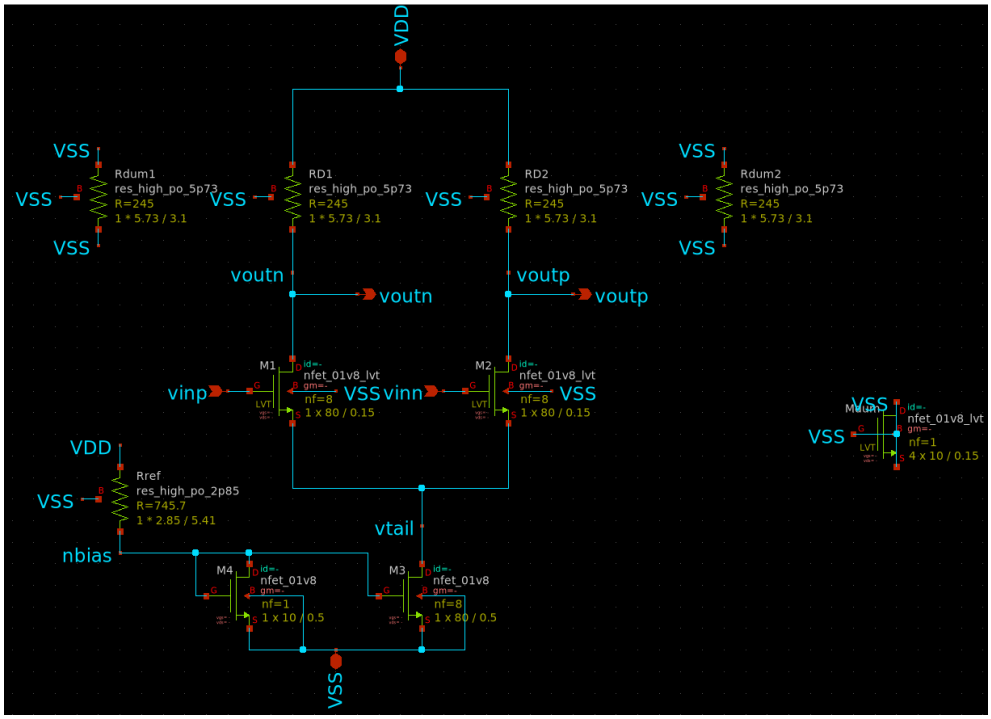


Figure 8: The adopted $W = 80\mu m$ BoMu schematic `ip/bomu.sch`: M_1/M_2 pair, `res_high_po` drain loads R_{D1}/R_{D2}, and the resistor-programmed N=8 tail mirror (M_3/M_4, R_{ref}).

Table 8: Integrated AC verification (`sim/tb_ac.spice`): BoMu with the real tail mirror ($W_{out} = 80$, $W_{ref} = 10$, $L = 0.5$, $N = 8$), sweeping the programming resistor. $W = 80\mu m$ pair, ideal $R_D = 245\Omega$, $V_{SC} = 1.052V$.

R_{ref} [Ω]	V_{tail} [V]	I_{tail} [mA]	A_{v0} [dB]	f_{-3dB} [MHz]
720	0.332	2.32	10.26	167.8
745	0.333	2.28	10.19	167.3
760	0.334	2.26	10.15	167.1

9 Large-Signal Swing and Distortion

The $400mV_{pp}$ differential output swing (Table 1) is a large-signal requirement, so it is checked with a dedicated transient testbench (`ip/tb_bomu_swing.sch`, Figure 9, which instantiates the `bomu.sym` symbol hierarchically). A differential $1MHz$ sine rides on the $V_{SC} = 1.052V$ input common mode with per-source amplitude $V_a = 32mV$, so $v_{id,pp} = 4V_a = 128mV_{pp}$ drives the amplifier to the swing target. At the `tt` corner ($1.2V$, 27°C) the measured swing is $v_{od,pp} = 400mV_{pp}$, on target, with each single-ended output spanning 0.824–$1.024V$ about a $\sim 0.924V$ common mode — inside the rails with every device still saturated, so the stage is not clipping. The same deck reports the AC gain/bandwidth and the averaged supply power ($A_{v0} = 10.1dB$, $f_{-3dB} = 168MHz$, $2.74mW$ pair / $3.10mW$ total including the reference leg), so swing, gain, bandwidth and power all come from one run.

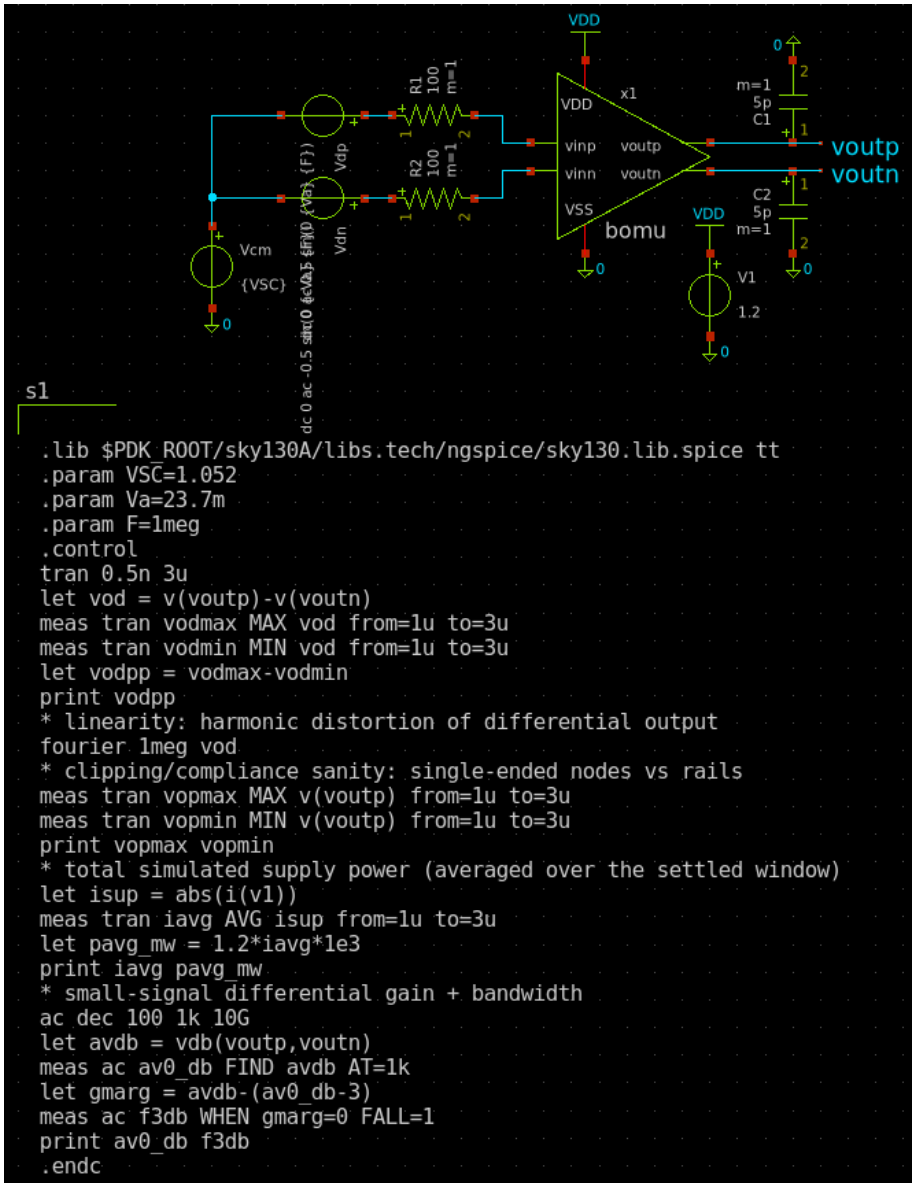


Figure 9: The transient swing testbench `ip/tb_bomu_swing.sch`.

Third-harmonic dominance. A Fourier decomposition of v_{od} (fundamental $1MHz$, ten harmonics) gives THD = 0.61%, set almost entirely by the third harmonic (Table 9). Two features are characteristic of a balanced source-coupled pair. First, the even harmonics vanish (at the $\sim 10^{-14}$ numerical floor): even-order distortion is common-mode and cancels in $v_{od} = v(\text{voutp}) - v(\text{voutn})$, so their absence confirms the devices and loads are symmetric. Second, the residual distortion is odd-order, led by the third. Both features follow from the *shape* of the balanced pair's differential transfer $i_{od}(v_{id})$, independently of any particular device law: it is an *odd* function of the differential drive (swapping the inputs negates the output), which is precisely what forces the even harmonics to cancel, and it is *compressive* (the pair steers a fixed tail current, so i_{od} saturates at $\pm I_{tail}$). An odd, compressive characteristic has a Taylor expansion in odd powers of v_{id} alone, so the leading nonlinearity is the cubic term,[1] which folds back as a third harmonic. The fifth and higher harmonics fall off by orders of magnitude, hence THD $\approx$ the third-harmonic term alone. Because that term scales with the cube of the drive, distortion climbs quickly with swing; at the $400mV_{pp}$ operating point it stays a clean 0.61%.

[1] There is no closed-form expression for $i_{od}(v_{id})$ at BoMu's operating point. The often-quoted $i_{od} \propto \tanh(v_{id}/2nV_T)$ is the *weak-inversion limit* only, where the devices follow the exponential law $i_D \propto \exp(v_{GS}/nV_T)$ ($V_T = kT/q$ the thermal voltage, n the subthreshold slope factor, highest g_m/I_D); the opposite strong-inversion limit is the square-law $\sqrt{}$-characteristic. BoMu runs at $g_m/I_D \approx 15$ (moderate inversion), between the two, where neither closed form holds — but the odd-symmetry and leading-cubic arguments above hold throughout.

Table 9: Fourier spectrum of v_{od} at the $400mV_{pp}$ operating point (`tt`, $1.2V$, 27°C, $1MHz$ drive).

Harmonic	Freq [MHz]	Norm. mag	
1	1	1.000	fundamental ($0.201\,V$ ampl.)
2	2	4×10^{-14}	even (cancelled)
3	3	6.1×10^{-3}	dominant ($\approx$ THD)
4	4	5×10^{-14}	even (cancelled)
5	5	7.0×10^{-6}	

10 Small-Signal Quality: Offset, CMRR, PSRR and Noise

Beyond the headline gain/bandwidth/swing targets, the pair is characterized for input offset, common-mode and supply rejection, and noise, at `tt`/$1.20V$/ 27°C. These figures are measured on the *extracted, post-layout* netlist (`verify/bomu_rcx.spice`, full $R + C$ RCX of the signed-off W=80μm layout) via `sim/corners.py acq rcx` and `sim/corners.py mc_mm --pex rcx`, so the numbers below carry the real device sizing *and* the layout parasitics. One point about the models frames every mismatch-dependent number. Contrary to a common belief about the open PDK, sky130A *does* carry native statistical device mismatch: the `.pm3` model cards apply per-device Gaussian spread to V_{th0}, t_{oxe}, v_{off} and n_{factor} through `mc_mm_switch`·`AGAUSS`(...) terms, live in `nfet_01v8_lvt` (the pair), `nfet_01v8` (the mirror) and `res_high_po` (the loads). Because the extracted `.subckt` wraps these same sky130 device instances, the native mismatch still redraws per device inside the parasitic netlist. The offset below is therefore drawn from this native mismatch with *no* assumed Pelgrom coefficient (`mc_mm_switch` flipped via `alterparam` after the `.lib` loads, then re-seeded per draw; see `corners.py mc_mm --pex`).

Input offset. A 40-draw native-mismatch Monte Carlo on the extracted W=80μm pair ($12\mu m^2$ per device) gives $\sigma(V_{os}) \approx 2.8mV$ ($3\sigma \approx 8.3mV$, worst draw $7.4mV$), centred on zero (mean $+0.6mV$). Because offset grows as $1/\sqrt{WL}$, this is set by the pair size the gain–bandwidth sizing already demands; it is modest by construction and protected by the common-centroid layout of Section 11 — indeed the extracted interdigitated pair sits a little below the $\sim 3.6mV$ a single lumped W=80μm device predicts. Offset is treated as a separate, un-targeted spec here.

CMRR and PSRR. Because the output is taken differentially between matched halves, common-mode input and supply ripple couple almost entirely into the output common mode and cancel differentially. The differential CMRR and PSRR are matching-limited, not tail-limited, and would be infinite for a perfectly symmetric netlist; over the native-mismatch draws on the extracted netlist their median is CMRR $\approx 67dB$ and PSRR $\approx 53dB$ (both scale inversely with the per-draw mismatch, so they are quoted as the median of the 24-draw set). The underlying common-mode paths set the ceiling and are finite: the tail-limited common-mode gain is $A_{cc} = -17.6dB$, and because R_D couples V_{DD} straight onto each drain — now reinforced by the asymmetric V_{DD}-to-drain coupling capacitors the extraction adds — the supply-to-common gain rises to $A_{sc} = -7.8dB$. This is the same balance that suppressed the even harmonics in Section 9.

Noise. The input-referred noise of the extracted netlist integrated over $10Hz$–$1GHz$ is $\approx 93\mu V_{rms}$ (device thermal/flicker dominated, so it is taken at the nominal matched point rather than per mismatch draw). Against the $\sim 34mV_{rms}$ input drive that produces the $400mV_{pp}$ output swing, this is an input SNR of order $51dB$. The integration band runs well past the $\sim 164MHz$ output pole, so the figure is conservative: above the pole the falling gain inflates the input-referred density, and most of the integral accumulates beyond the usable band.

Table 10: Post-layout small-signal quality on the extracted netlist (`verify/bomu_rcx.spice`) at `tt`/$1.20V$/27°C. Input offset is a 40-draw native sky130A device-mismatch Monte Carlo (no assumed A_{VT}); the differential CMRR/PSRR are mismatch-limited and quoted as the median of a 24-draw set (they scale inversely with the per-draw mismatch). Noise is at the nominal matched point.

Metric	**Value**	**Condition**
Input offset $\sigma(V_{os})$	≈ 2.8 mV	native-mismatch MC (40 draws)
Input offset 3σ	≈ 8.3 mV	native-mismatch MC (40 draws)
CMRR (differential)	≈ 67 dB	native mismatch (median, 24 draws)
PSRR (differential)	≈ 53 dB	native mismatch (median, 24 draws)
Common-mode gain A_{cc}	-17.6 dB	tail-limited
Supply→CM gain A_{sc}	-7.8 dB	R_D + parasitic coupling
Input-referred noise	$\approx 93\mu V_{rms}$	$10Hz$–$1GHz$, nominal

11 Physical Layout in `magic`

With the $W = 80\mu m$ design signed off against the `tt` corner, the `bomu` cell is drawn in `magic` on SKY130A from the adopted schematic (Table 7), following the matching and isolation methodology below.

Pair core. The overriding layout concern for a differential pair is matching of M_1/M_2, so the core is a common-centroid interdigitation of the $nf = 8$ pair. The $W = 80\mu m/nf = 8$ pair lays out as $10\mu m$-per-finger unit devices on isolated diffusion islands, assigned to the two halves in an alternating centroid pattern so a linear process gradient across the block cancels differentially.

Edge dummies. Matching is protected at the array edges by same-width $nf = 1$ dummy transistors on the left and right columns, which give the outermost live fingers an identical poly/diffusion neighbourhood ($10\mu m$ finger width, matching the pair). Their gates, sources and drains are strapped to `VSS`. The load resistors are handled the same way: the two live loads R_{D1}/R_{D2} (`res_high_po_5p73`) are flanked by two dummy resistors of identical geometry for etch/density symmetry, terminated to `VSS`.

Tail mirror and guard rings. The mirror lays out as a single interdigitated row of `nfet_01v8` unit fingers ($W = 10\mu m$, $L = 0.5\mu m$) with the diode-connected reference M_4 at the centre and the eight M_3 output fingers around it (a 1-D common-centroid, ratio $N = 8$); the programming resistor R_{ref} (`res_high_po_2p85`) sits beside it. Being an all-NMOS block, every sub-block is wrapped in a p^+ guard ring tied to `VSS` with no n-well, and the six top-level ports (`vinp`, `vinn`, `voutp`, `voutn`, `VDD`, `VSS`) are brought out on upper metal past the outer ring. The finished cell is shown in Figure 10.

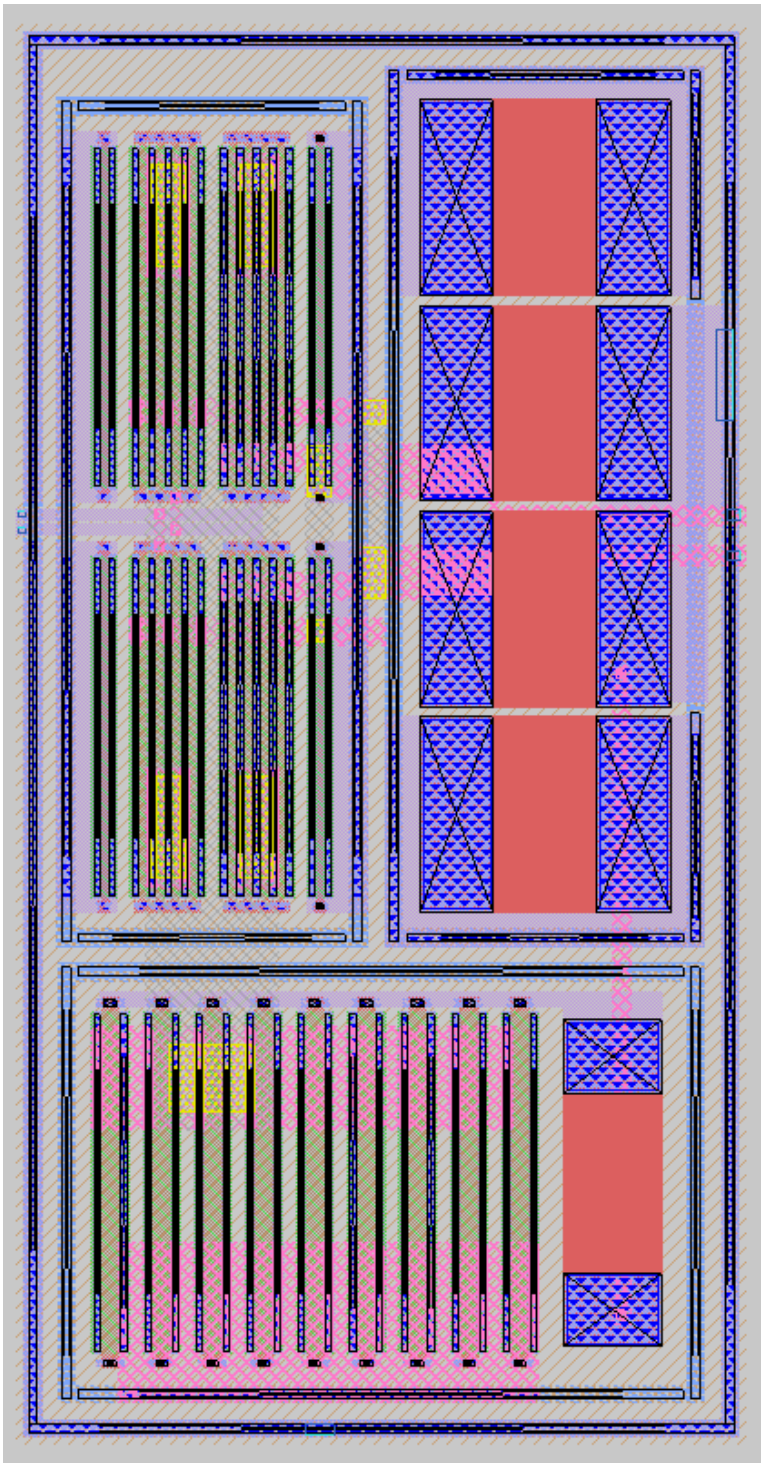


Figure 10: The signed-off `bomu` cell in `magic` (SKY130A): the common-centroid $W = 80\mu m$ pair with edge dummies, the dummy-flanked `res_high_po` load bank, and the interdigitated tail mirror, each wrapped in a `VSS`-tied p^+ guard ring.

Verification and sign-off. The block closes LVS (`netgen` against `ip/bomu.sch`) *match uniquely*, and both DRC decks — `magic` geometric and the `klayout sky130A_mr` sign-off runset on the written GDS — report zero violations. Parasitics are then extracted (full $R+C$ RCX via `magic`'s `extresist`) and the `tt` AC response re-run against the extracted netlist. Table 11 compares the schematic and post-layout operating point: as Section 8.7 anticipated, the parasitics load the output pole (a few-percent bandwidth shave) but leave the DC gain essentially untouched, and the $\sim 12\%$ pre-layout bandwidth margin absorbs the shift — the extracted design still clears both the $10dB$ gain and $150MHz$ bandwidth targets.

Table 11: Adopted $W = 80\mu m$ design: schematic vs. post-layout (RCX, full $R + C$) at the `tt` corner, 27°C, $V_{DD} = 1.20V$. Both specs ($A_{v0} \geq 10dB$, $f_{-3dB} \geq 150MHz$) are met with parasitics.

Quantity	**Schematic**	**Post-layout (RCX)**	**Unit**
Differential gain A_{v0}	10.1	10.19	dB
Bandwidth f_{-3dB}	168	164.1	MHz

The signed-off GDS and the LVS/DRC reports are archived under `tapeout/` and `verify/`; the extraction and compare scripts are `verify/ext_rcx.tcl`, `verify/run_lvs.sh` and `verify/drc.sh`, and the post-layout corner sweep is `sim/corners.py pex rcx`.

12 Status and Next Steps

The central result (Section 8) is that the $10dB$ / $150MHz$ specification sits on the single-stage gain–bandwidth wall $g_m/(2\pi C_L)$, which the baseline sizing clears by only $\sim 1.5\%$. A small g_m increase and a parasitic margin give the **adopted** $W = 80\mu m$ **design** ($R_D \approx 245\Omega$, $I_{tail} = 2.28mA$, $2.74mW$), which meets both targets at the `tt` corner — $A_{v0} = 10.1dB$, $f_{-3dB} = 168MHz$, $400mV_{pp}$ swing at 0.61% THD — and holds through the native device-mismatch MC (Section 10), which moves offset but not the differential gain/bandwidth. On the area–power plane (Sections 8.3–8.4) it sits near the joint minimum of area *and* power the `tt` criteria allow, on the $g_m/I_D = 15$ ray that is the optimum of a power-weighted cost.

The physical implementation (Section 11) is now complete and **signed off**: the $W = 80\mu m$ common-centroid pair, the dummy-flanked load bank and the interdigitated tail mirror are drawn with `VSS`-tied guard rings, the block closes `netgen` LVS *match uniquely* and passes both the `magic` and `klayout` `sky130A_mr` DRC decks with zero violations, and the extracted-parasitic (RCX) `tt` AC re-simulation confirms the $\sim 12\%$ bandwidth margin absorbs the parasitic pole shift with the gain intact (Table 11). The signed-off GDS and reports are archived under `tapeout/` and `verify/`.

The offset/CMRR/PSRR/noise characterisation of Section 10 is now run directly on the extracted (post-layout) netlist: the native device-mismatch Monte Carlo and the CMRR/PSRR/noise measurements all exercise `verify/bomu_rcx.spice`, so they carry both the adopted W=$80\mu m$ sizing and the layout parasitics. The headline differential gain/bandwidth are untouched by mismatch (as expected for a balanced pair); the quality metrics land at $\sigma(V_{os}) \approx 2.8mV$, CMRR $\approx 67dB$, PSRR $\approx 53dB$ and $\approx 93\mu V_{rms}$ input-referred noise, with the supply-rejection path the most parasitic-sensitive (A_{sc} rising from the ideal R_D-coupling estimate once the extracted V_{DD}-to-drain capacitors are included).

References

[1] P. G. A. Jespers and B. Murmann, *Systematic Design of Analog CMOS Circuits Using Pre-Computed Lookup Tables*. Cambridge University Press, 2017.

[2] B. Razavi, *Design of Analog CMOS Integrated Circuits*, 2nd ed. McGraw-Hill, 2017, ch. 4: Differential Amplifiers.